

7. The heart is a pump which supplies blood to the organs of the body. It deals with increasing physical activity, such as exercise, by increasing its rate (number of beats per minute). After exercise, the heart rate will eventually return to the resting rate. The time the heart takes to return to its resting rate is a good indication of a person's level of fitness.

A researcher wanted to investigate the difference in the level of fitness between men and women. Six people at a gym volunteered to compare their fitness. Each volunteer ran on a treadmill for five minutes. Immediately at the end of the exercise their heart rates were recorded on a laptop computer. Their heart rates were also recorded one minute later.



The researcher conducting the investigation used the data collected from each volunteer to calculate their fitness index. An index of 60 – 70 shows a person who is reasonably fit, below this level shows decreasing fitness.

The results of the investigation are shown in the table below.

Volunteer	Resting heart rate (beats/min)	Heart rate immediately after exercise (beats/min)	Heart rate 1 min after exercise (beats/min)	Fitness index	Body mass (kg)
John	70	190	130	35	56.5
Michael	70	150	80	60	63.6
Charlotte	60	110	65	75	76.4
Paul	80	195	140	20	101.8
Peter	75	180	156	43	59.1
Lucy	70	130	80	68	56.1

(a) (i) Who appears to be the least fit person in the table? Explain your answer. [1]

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Examiner
only

(ii) What is the relationship between **fitness index** and heart rate immediately after exercise? [1]

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(iii) At the end of the investigation the volunteers discussed the results. Peter said that “he thought that John was fitter than Lucy because one minute after exercise John’s heart rate had fallen by 60 beats/min whereas Lucy’s had only fallen by 50 beats/min.” The researcher said that you must calculate the percentage decrease in the heart rates in order to make a fair comparison.

I. In the space below calculate the percentage decrease in the heart rates of both John and Lucy. [2]

John

Percentage decrease in heart rate =

Lucy

Percentage decrease in heart rate =

II. From the answers to (I.) above state whether John or Lucy appears fittest. [1]

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(b) The design of this investigation has many faults.

Give **two** factors which should be controlled in this investigation. [1]

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(c) Why is it not possible to conclude from the data that Paul’s low **fitness index** is due to obesity? [1]

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